

A Deterministic, CVSS–Environmental Method for FedRAMP Rev5 VDR/VER Vulnerability Prioritization

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Draft Whitepaper / Request for Comments

June 2026 | Status: Informational Draft

Abstract

FedRAMP Rev5 introduces a Vulnerability Disclosure Report (VDR) and Vulnerability Evaluation Requirements (VER) that define *Potential Agency Impact* (PAIN, levels N1–N5) and a remediation–timeframe matrix, together with evaluation rules for likely exploitability and internet reachability. FedRAMP deliberately does *not* prescribe how to derive a PAIN level from an individual vulnerability on a specific asset. This memo argues that the missing classifier already exists, in standardized form, as the **Environmental metric group of CVSS**: the Confidentiality, Integrity and Availability *Requirements* (CR/IR/AR) and the Modified Impact Sub-Score were designed precisely to re-weight a vulnerability’s impact by what is at stake in a given deployment. We present a closed-form, reproducible mapping from FedRAMP’s requirements onto CVSS Environmental metrics, derive PAIN and the remediation deadline from it, and give fully worked calculations. We also present an *example* mechanism — “asset archetypes” — for assigning CR/IR/AR systematically, while emphasizing that each Cloud Service Provider (CSP) SHOULD derive its own assignment from its system categorization. A reference architecture and a sample scan complete the memo.

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1 Status of This Memo and Terminology

This document is an informational draft circulated for comment. It does not define a FedRAMP requirement; it proposes a standardized *method* for satisfying existing VDR/VER requirements.

The key words MUST, MUST NOT, SHOULD, SHOULD NOT, and MAY are to be interpreted as described in RFC 2119.

PAIN

Potential Agency Impact, the FedRAMP N1–N5 rating of a finding.

CR/IR/AR

The CVSS Environmental Confidentiality / Integrity / Availability *Requirements* of an asset.

ISC

Impact Sub-Score (here, the CVSS *Modified* Impact Sub-Score).

LEV / NLEV

Likely Exploitable / Not Likely Exploitable (a VER axis).

IRV / NIRV

Internet Reachable / Not Internet Reachable (a VER axis).

Class

The provider Certification Class (A–D) selecting a remediation table.

2 Motivation: the gap between VDR/VER and the scanner

IN PLAIN TERMS

A scanner tells you a flaw exists and how severe it is *in the abstract*. FedRAMP asks a different question: how bad is this flaw *for the agency, on this exact system*, and how fast must we fix it? A “low”-rated flaw on the database holding everyone’s records can matter more than a “high” on a throwaway test box. This memo gives one consistent, repeatable way to answer FedRAMP’s question — instead of leaving it to each analyst’s judgment.

A vulnerability scanner emits, per finding, a CVE identifier, a qualitative severity, and (often) a CVSS Base vector. FedRAMP Rev5 VDR/VER, by contrast, asks the provider to answer a different question: *what is the potential impact on the agency if this vulnerability is exploited on this asset, and how quickly must it therefore be remediated?* The two are not the same. A “Low”-rated CVE on a crown-jewel datastore can carry more agency impact than a “High”-rated CVE on a throwaway sandbox.

FedRAMP supplies the *output* vocabulary (the PAIN buckets and the remediation matrix) and the *evaluation axes* (likely-exploitable, internet-reachable) but leaves the *classifier* — the function from (CVE, asset) to PAIN — to the provider. An ad-hoc or analyst-by-analyst classifier is neither reproducible nor defensible. This memo supplies a deterministic one.

3 Key insight: FedRAMP’s requirements are CVSS Environmental metrics

IN PLAIN TERMS

You don’t need a brand-new risk formula. The industry-standard CVSS scoring system already has a built-in way to say “this system is especially critical, so weight its flaws more heavily.” That built-in knob is exactly what FedRAMP’s “impact on the agency” is asking for. We simply turn the knob on, instead of inventing something new.

The central claim of this memo is that FedRAMP’s notion of “impact on the agency” maps, almost one-to-one, onto metrics that already exist in the CVSS v3.1 specification’s *Environmental* group. The Environmental group exists for exactly this purpose: to let the consuming organization re-score a vulnerability according to the criticality of the affected asset in *its* environment.

FedRAMP VDR/VER concept	CVSS mechanism that satisfies it
Asset data sensitivity / FIPS-199 C-I-A categorization	Security Requirements CR, IR, AR
Magnitude of potential agency impact	Modified Impact Sub-Score (ISC)
Likely exploitable (VER)	LEV: exploit maturity (CISA Vulnrichment) / EPSS threshold
Internet reachable (VER)	IRV: deployment/network context
Single- vs. multi-agency blast radius	scope multiplier on the PAIN tier
Provider Certification Class	selector of the remediation deadline table

Table 1: FedRAMP requirements fall directly onto CVSS Environmental metrics.

The consequence is that a conforming implementation need not invent a bespoke risk algebra. It need only (a) assign each asset its CR/IR/AR, and (b) evaluate the two VER axes. The arithmetic that turns those inputs into a defensible impact magnitude is the published CVSS Environmental formula.

4 The model

IN PLAIN TERMS

Two questions decide everything. (1) **How much damage** could this flaw do to *this* system? (2) Does the system hold **one agency's** data or **many**? Question 1 produces a word — Minimal, Narrow, Disruptive, or Debilitating. Question 2 decides whether a “Disruptive” or “Debilitating” result counts once or is escalated because several agencies are exposed. Together they pick the final N1–N5 level.

We define PAIN as a function of a *severity* term and a *scope* term:

$$\text{PAIN}(\text{finding}) = f(\text{SEVERITY}, \text{SCOPE}).$$

SEVERITY answers “how badly does this vulnerability’s impact land on *this* asset?” — the CVE’s impact metrics re-weighted by the asset’s CR/IR/AR. **SCOPE** answers “how many agencies’ data does the asset hold?” — single (1) or multiple (> 1). SEVERITY is mapped to a FedRAMP customer-effect word (Minimal, Narrow, Disruptive, Debilitating); the word and scope together select the N-level.

5 Two threat inputs from CISA Vulnrichment

IN PLAIN TERMS

Beyond the CVE’s own CVSS vector, the method pulls in two facts about each flaw from *CISA Vulnrichment* — CISA’s Authorized Data Publisher (ADP) enrichment of CVE records: how the flaw is actually being exploited, and how much control exploitation grants. One tightens the remediation *clock*; the other can raise the PAIN *level*. Both are optional: without them the method falls back to the CVSS vector and EPSS alone.

Two distinct inputs feed two distinct stages:

- **Exploitation state** (active / PoC / none) → the **remediation clock**. “active” makes a finding Likely Exploitable (LEV) regardless of its EPSS probability, moving it to a faster remediation column (Section 8).
- **Technical impact** (total / partial) → the **PAIN level**. “total” floors the in-scope CVSS impact to High before the CR/IR/AR weighting, which can push a finding up a tier (Section 6).

How it plays in — three quick illustrations (Class C, single-agency):

- *Exploitation moves the clock*. An N3 finding with EPSS 0.30 (below threshold) sits in the slow column: **128 days**. Mark it **exploitation = active** and it becomes LEV — the deadline drops to **16 days** (internet-reachable) or 32 days (not).
- *Technical impact moves the level*. A “weak” RCE (low impact across C/I/A) on a High/High-/High asset scores Disruptive → **N3**. Mark it **technical impact = total** and the in-scope dimensions floor to High, scoring Debilitating → **N4** (worked in Example 5).

- *Neither present.* No active exploitation, sub-threshold EPSS, and partial/absent technical impact → the finding is scored purely on its CVSS vector and stays in the slow column at its base tier.

6 The mathematics

IN PLAIN TERMS

This is the engine room — you can skim the equations and still follow the idea. Every flaw can hurt three things: **secrecy**, **correctness**, and **uptime**. For each, we multiply “how badly the flaw breaks it” by “how much this system depends on it,” then combine the three into a single damage score between 0 and 1. Finally we translate that score into a plain word. Bigger number = more damage. The only place human judgment enters is the three cut-points that decide where one word becomes the next.

All constants below are taken *verbatim* from the CVSS v3.1 specification; only the normalization and the word thresholds (Section 6) are additions.

IN PLAIN TERMS

What the symbols mean.

- C, I, A — how badly *this flaw* breaks secrecy, correctness, and uptime (from the CVE’s CVSS vector). None / Low / High become 0 / 0.22 / 0.56.
- CR, IR, AR — how much *this system* needs secrecy, correctness, and uptime (from its archetype). Low / Medium / High become 0.5 / 1.0 / 1.5.
- ISC — the three combined into one “damage to this system” number.
- S — that damage rescaled to a clean 0–1 range.
- W — the plain English word S maps to.
- m — does the system serve one agency (0) or many (1)? Resolved hierarchically from the workload-/namespace/cluster tags.
- $\theta_1, \theta_2, \theta_3$ — the three cut-points where the word changes.

Impact metric weights. Each of the affected vulnerability’s Confidentiality, Integrity and Availability impact metrics (C, I, A), read from the CVSS Base vector, takes a weight:

$$C, I, A \in \{ \text{None} = 0, \quad \text{Low} = 0.22, \quad \text{High} = 0.56 \}.$$

Security Requirement weights. Each asset’s Requirement (CR, IR, AR) takes a multiplier:

$$\text{CR, IR, AR} \in \{ \text{Low} = 0.5, \quad \text{Medium} = 1.0, \quad \text{High} = 1.5 \}.$$

Modified Impact Sub-Score. The per-dimension impacts and requirements combine as a probabilistic union — the share of the asset left uncompromised on each dimension is multiplied, and the complement is the overall impact:

$$\text{ISC} = \min \left[1 - (1 - C \cdot \text{CR})(1 - I \cdot \text{IR})(1 - A \cdot \text{AR}), \quad 0.915 \right]. \quad (1)$$

The cap $0.915 = 1 - (1 - 0.56)^3$ is the CVSS ceiling (all-High impact at Medium requirements). We then normalize to a unit scalar:

$$S = \frac{\text{ISC}}{0.915} \in [0, 1]. \quad (2)$$

Severity word. S is bucketed into a FedRAMP customer-effect word by three cut points $(\theta_1, \theta_2, \theta_3)$:

$$W(S) = \begin{cases} \text{Minimal} & S < \theta_1 \\ \text{Narrow} & \theta_1 \leq S < \theta_2 \\ \text{Disruptive} & \theta_2 \leq S < \theta_3 \\ \text{Debilitating} & S \geq \theta_3 \end{cases} \quad (\theta_1, \theta_2, \theta_3) = (0.25, 0.55, 0.80). \quad (3)$$

The cut points are the model’s *one* calibratable judgment. They SHOULD be documented and back-tested against analyst-rated findings, and an implementation MUST treat them as governed configuration rather than an in-band, ad-hoc setting.

Scope and the N-level. Let $m \in \{0, 1\}$ be the asset’s multi-agency flag, resolved most-specific-first: workload label, then namespace label, then a namespace glob, then the cluster default (Section 10). Scope is purely hierarchical — an asset is multi-agency only if it (or its namespace, or the cluster) is tagged so. The N-level follows from the word and scope:

$$\text{PAIN} = \begin{cases} \text{N1} & W = \text{Minimal} \\ \text{N2} & W = \text{Narrow} \\ \text{N3} & W = \text{Disruptive}, m = 0 \\ \text{N4} & W = \text{Disruptive}, m = 1 \text{ or } W = \text{Debilitating}, m = 0 \\ \text{N5} & W = \text{Debilitating}, m = 1 \end{cases} \quad (4)$$

Technical-impact floor (optional refinement). Where an authoritative source (e.g. CISA Vulnrchment SSVC) provides a *technical impact* of **total**, the impacts that the CVE *does* touch may be floored to High before Eq. (1):

$$\hat{X} = \begin{cases} 0.56 & X > 0 \text{ and tech. impact} = \text{total} \\ X & \text{otherwise} \end{cases} \quad X \in \{C, I, A\}.$$

Crucially the floor MUST NOT invent impact on a dimension the CVE marks None ($X = 0$ stays 0); **partial** or **absent** leaves the vector unchanged.

7 Fail-safe behavior

IN PLAIN TERMS

If a system hasn’t been classified yet, we assume the worst, not the best — it scores high and shows up loudly until someone classifies it. Silence never makes a problem look smaller than it is.

Missing metadata MUST NOT *lower* PAIN. An asset that carries no classification SHOULD resolve to a deliberately conservative default (e.g. CR/IR/AR all High), which scores loudly and surfaces the asset for classification rather than hiding it. “Unknown” is treated as “serious until proven otherwise.”

8 Remediation: the VDR-TFR-PVR matrix

IN PLAIN TERMS

The N-level says *how bad*; this step says *how fast*. Two real-world facts tighten the clock: is the flaw actually being exploited (or very likely to be), and is the system reachable from the internet? More severe + more exploitable + more exposed = less time to fix. A lookup table turns those into a concrete deadline, stricter for higher-assurance providers.

The remediation deadline is selected by three values: the provider Certification Class, the PAIN level, and an exploitability *column* derived from the two VER axes:

$$\text{deadline} = M[\text{Class}][\text{PAIN}][\text{column}], \quad \text{column} \in \{\text{LEV}+\text{IRV}, \text{LEV}+\text{NIRV}, \text{NLEV}\}.$$

where, for a provider-selected EPSS threshold θ_{epss} (this memo uses 0.70):

$$\text{LEV} = (\text{EPSS} \geq \theta_{\text{epss}}) \vee (\text{exploitation} = \text{active}), \quad \text{IRV} = \text{asset is internet-reachable}.$$

Here *internet-reachable* means the asset is fronted by a **public-facing load balancer or route whose edge does not itself enforce authentication**. A route gated by native edge authentication — an identity-aware proxy or equivalent that authenticates every request before it reaches the workload — is treated as *not* internet-reachable (NIRV), on the view that an unauthenticated attacker cannot reach the vulnerable surface. (How a given platform's routes and edge authentication are discovered is an implementation concern, out of scope here.)

Both the exploitation state here and the technical impact in Section 6 are sourced from *CISA Vulnrichment*, an Authorized Data Publisher (ADP) in the CVE Program; that choice is itself a design opinion (see Appendix C). The day-valued tables are given in Appendix B. PAIN-1 carries no FedRAMP deadline.

9 Worked examples

IN PLAIN TERMS

The same formula, run by hand on realistic cases. Watch the *same* vulnerability land at very different N-levels depending only on which system it sits on — that is the entire point of the method.

We use C, I, A for impacts and $\text{CR}/\text{IR}/\text{AR}$ for requirements; all arithmetic follows Eqs. (1)–(4).

Example 1 — RCE on a crown-jewel datastore, multi-agency. $C = I = A = 0.56$ (High), $\text{CR}=\text{IR}=\text{AR}=1.5$ (High), $m = 1$.

$$\text{ISC} = \min[1 - (1 - 0.84)^3, 0.915] = \min[0.9959, 0.915] = 0.915,$$

$$S = 0.915/0.915 = 1.000 \Rightarrow W = \text{Debilitating},$$

$$m = 1 \Rightarrow \boxed{\text{N5}}.$$

Example 2 — the same CVE on a sandbox, single-agency. $C = I = A = 0.56$, $\text{CR}=\text{IR}=\text{AR}=0.5$ (Low), $m = 0$.

$$\text{ISC} = 1 - (1 - 0.28)^3 = 1 - 0.72^3 = 0.6268,$$

$$S = 0.6268/0.915 = 0.685 \Rightarrow \text{Disruptive} \Rightarrow \boxed{\text{N3}}.$$

Identical vulnerability, opposite ends of the matrix — the asset half of the score ($\text{CR}/\text{IR}/\text{AR}$) is doing the work.

Example 3 — availability-only DoS on a message backbone. A CVE with $C = 0$, $I = 0$, $A = 0.56$ (e.g. a parser DoS), on an asset with $CR=IR=AR=1.5$, $m = 0$. $EPSS = 0.76$, not internet-reachable.

$$ISC = 1 - (1 - 0)(1 - 0)(1 - 0.84) = 1 - 0.16 = 0.84,$$

$$S = 0.84/0.915 = 0.918 \Rightarrow W = \text{Debilitating}, m = 0 \Rightarrow \boxed{N4}.$$

The single dimension $A \cdot AR = 0.84$ alone clears the Debilitating bar. Remediation: LEV ($EPSS \geq 0.70$) $\wedge$ NIRV; at Class C this is $M[C][N4][LEV+NIRV] = 8$ days. Note this finding may carry a *Low* vendor severity — PAIN is decoupled from the qualitative label.

Example 4 — information disclosure on a metadata-only service. $C=0.56$, $I=0$, $A=0$, on an asset with $CR=0.5$ (Low), $IR=AR=1.5$.

$$ISC = 1 - (1 - 0.28)(1)(1) = 0.28,$$

$$S = 0.28/0.915 = 0.306 \Rightarrow W = \text{Narrow} \Rightarrow \boxed{N2}.$$

A confidentiality leak on a service that holds only operational metadata is reconnaissance, not a debilitating breach — captured by the low CR.

Example 5 — the technical-impact floor. A “weak” RCE with $C = I = A = 0.22$ (Low) on a High/High/High asset, single-agency. Without the floor: $ISC = 1 - (1 - 0.33)^3 = 0.699$, $S = 0.764 \Rightarrow$ Disruptive $\Rightarrow$ N3. If an authoritative source reports technical impact **total**, the touched dimensions floor to 0.56, giving $S = 1.0 \Rightarrow$ Debilitating $\Rightarrow \boxed{N4}$.

10 Assigning CR/IR/AR: archetypes (an example, not a standard)

IN PLAIN TERMS

Rather than hand-rate every server, we sort systems into a short list of “types” — a database, a login service, a cache, a message bus, and so on — and each type comes with its criticality pre-filled. Tag a system with its type and the scoring is automatic. Our list is only an *example*: every provider should tailor its own to the data it actually holds.

The model in Section 6 is only as good as the CR/IR/AR assignment that feeds it. Assigning requirements per individual asset does not scale and invites drift. A practical mechanism is to define a small catalog of *asset archetypes*, each carrying a fixed CR/IR/AR, and to tag each asset with one archetype.

The archetype catalog in Appendix A is illustrative. A conforming implementation MUST derive its CR/IR/AR assignment from its own system categorization (FIPS-199 impact levels, data inventory, authorization boundary). It MAY adopt the example catalog as a starting point, but each CSP SHOULD own and justify its mapping. Two providers with different data inventories can legitimately assign the same software different requirements.

The classification rule we recommend is *control-plane lens first*: if an asset can deploy, orchestrate, hold cross-estate credentials, or actuate configuration, classify it by that control function regardless of the data it stores; otherwise classify by the data it holds. The same physical software lands in different archetypes by role — an in-memory store used as a cache is **app-tier**, the same software used as a session/token store is **identity-secrets**, and used as a job broker is **data-backbone**.

11 Determinism, governance, and defensibility

IN PLAIN TERMS

Because it is a fixed formula, two people scoring the same flaw get the same answer — and you can show your work to an auditor line by line. The only adjustable dials are locked down as managed settings, not left for anyone to change on a whim.

- **Determinism.** Given the same (CVE vector, archetype, VER axes), the method yields the same PAIN and deadline on every run and for every analyst.
- **Two human inputs only.** An asset needs exactly two tags (**asset-archetype**, **multi-agency**); everything else is derived.
- **Governed calibration.** The word thresholds (Eq. (3)) and the EPSS threshold are the only tunable knobs and MUST live in governed configuration, not in ad-hoc cluster state.
- **Auditability.** Each finding can carry the full derivation: the resolved archetype and its source, CR/IR/AR, *S*, the word, the scope, and the selected matrix cell.

12 Reference architectures and sample scans

IN PLAIN TERMS

Two end-to-end pictures: a typical single-agency system, and a stricter shared system serving several agencies. Each box is tagged with its type; the tables show what the scan would report. Notice how the shared, multi-agency, high-assurance system earns far tighter deadlines — hours, not months — for the same kinds of flaws.

12.1 Single-agency, Class C

Figure 1 shows a representative deployment with each component tagged by archetype. Table 2 shows the corresponding per-finding output for a fictional Class C, single-agency offering. The PAIN column is computed by the method above; the Classical Severity column is the scanner’s qualitative rating, shown to make the decoupling explicit.

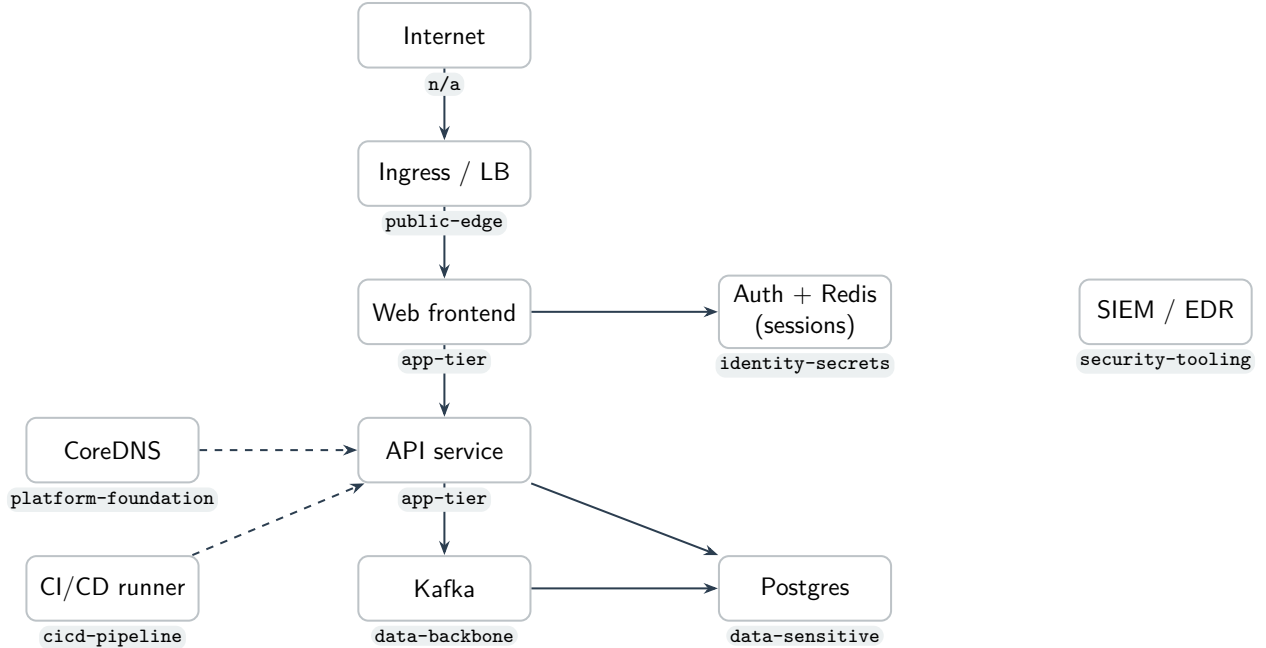


Figure 1: Reference architecture; each node carries its `asset-archetype` tag.

CVE	Resource (archetype)	Classical	EPSS	Internet	PAIN	Remediation
CVE-2026-1001	ingress (public-edge)	CRITICAL	0.97	yes	N4	4 days
CVE-2026-1002	payments-db (data-sensitive)	CRITICAL	0.94	no	N4	8 days
CVE-2026-1003	redis (identity-secrets)	HIGH	0.50	no	N4	64 days
CVE-2026-1004	kafka (data-backbone)	LOW	0.76	no	N4	8 days
CVE-2026-1005	web (app-tier)	MEDIUM	0.20	yes	N3	128 days
CVE-2026-1006	coredns (platform-foundation)	MEDIUM	0.08	no	N2	192 days

Table 2: Sample VDR scan output (fictional data; Class C, single-agency). PAIN is computed from the archetype’s CR/IR/AR and the CVE impact vector; the deadline is $M[C][PAIN][column]$. Note CVE-2026-1004: a *Low* vendor severity still yields N4 because it is an availability-High DoS on a High-availability-requirement asset (Example 3).

12.2 Multi-agency, Class D

Figure 2 is a shared, multi-tenant platform serving *several* agencies at the highest assurance (Class D). Its shared components carry more than one agency’s data, so they are tagged multi-agency (here via the cluster default, with single-agency tenant workloads tagged as the exceptions). Table 3 shows the result: the same families of flaws now remediate in *hours*.

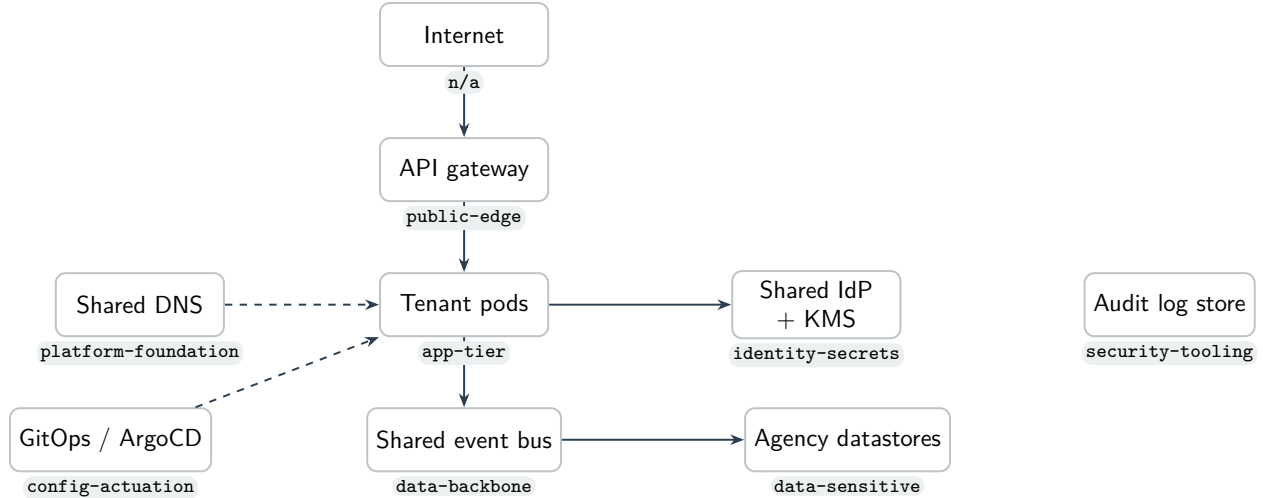


Figure 2: Multi-agency reference architecture; shared components serve several agencies and are tagged multi-agency.

CVE	Resource (archetype)	Classical	EPSS	Internet	PAIN	Remediation
CVE-2026-2001	gateway (public-edge)	CRITICAL	0.90	yes	N5	12 hours
CVE-2026-2002	shared-idp (identity-secrets)	CRITICAL	0.95	no	N5	1 day
CVE-2026-2003	event-bus (data-backbone)	LOW	0.80	no	N5	1 day
CVE-2026-2004	agency-db (data-sensitive)	HIGH	0.40	no	N5	8 days
CVE-2026-2005	tenant-pod (app-tier)	MEDIUM	0.10	no	N3	64 days
CVE-2026-2006	shared-dns (platform-foundation)	MEDIUM	0.05	no	N2	192 days

Table 3: Sample VDR scan output (fictional data; Class D, multi-agency). Multi-agency scope escalates Disruptive/Debilitating findings, and Class D carries the tightest deadlines — an internet-facing exploitable flaw on the shared edge remediates in **12 hours** (CVE-2026-2001). CVE-2026-2003 again shows a *Low* vendor severity at N5 (availability DoS on the shared backbone). A confidentiality leak on shared DNS (CVE-2026-2006) stays N2 — “Narrow” does not escalate with scope.

13 Relationship to CVSS v4.0

IN PLAIN TERMS

CVSS has a newer version (4.0) that changes how a flaw’s impact is described and scored. We build on the older 3.1 because it exposes the clean, transparent formula this method depends on. Findings that only have a 4.0 score can still be handled with a reasonable approximation, and in practice most published scores are still 3.1 today.

This method is expressed in CVSS v3.1 terms because the v3.1 Environmental group provides a closed-form Modified Impact Sub-Score (Eq. (1)) with explicit CR/IR/AR multipliers. CVSS v4.0 reworks impact into Vulnerable-System (VC/VI/VA) and Subsequent-System (SC/SI/SA) metrics, removes the **Scope** metric, and scores via a lookup “MacroVector” rather than a formula; CR/IR/AR persist but feed that lookup. A v4.0-only finding can be *approximated* by mapping

VC/VI/VA onto $C/I/A$ in Eq. (1), at the cost of ignoring Subsequent-System impact. While NVD remains v3.1-dominant, an implementation SHOULD prefer the v3.1 vector and MAY treat v4.0 via this approximation until a native v4.0 environmental scorer is warranted.

14 Security Considerations

IN PLAIN TERMS

The method trusts the type-tags on each system. If someone mislabels a critical system as trivial, its flaws will look smaller than they really are — so the tags must be protected like any other security setting.

The method’s integrity depends on the trustworthiness of two inputs: the asset archetype tags and the VER axes (EPSS / exploitation / reachability). Tag spoofing that *lowers* an archetype is a downgrade attack; the fail-safe default and governed calibration mitigate accidental under-classification, but tag assignment SHOULD be controlled like any other security configuration. The method does not detect vulnerabilities; it prioritizes the output of a scanner and inherits that scanner’s coverage and accuracy.

15 Proposal: a finer exploitability gradation (VLEV) for FedRAMP

IN PLAIN TERMS

This section is forward-looking — a proposal we would bring to FedRAMP, not part of the method as implemented today. Right now “exploitability” is a single yes/no line. We suggest splitting it into three bands so the very tightest remediation clocks are reserved for the flaws most likely to actually be used in the wild.

The added inflection point. The current model uses one EPSS cut to split Likely-Exploitable (LEV) from Not (NLEV). We propose *two* cuts and three bands:

- **NLEV** — $EPSS \leq 0.30$: not likely exploitable.
- **LEV** — $0.30 < EPSS \leq 0.75$: likely exploitable.
- **VLEV** — $EPSS > 0.75$: *very* likely exploitable.

Observed active exploitation always maps to VLEV, regardless of EPSS: if CISA Vulnrichment reports `exploitation = active` (KEV-equivalent), the flaw is being used *now* and belongs in the most-urgent band by definition.

The payoff is twofold. The 0.30–0.75 band — today swept into the slow NLEV column — carries real exploitation probability and earns a tighter clock than genuinely quiet vulnerabilities (≤ 0.30); and the > 0.75 / actively-exploited flaws earn a clock tighter than today’s single LEV tier provides.

Effect on the timelines, by Class. VLEV introduces a new tightest column — roughly half the current LEV+IRV deadline, floored at 12 hours. The existing LEV deadlines carry over to the new 0.30–0.75 band, and NLEV (now ≤ 0.30) keeps its relaxed deadline. The proposed VLEV+IRV floor:

PAIN	Proposed VLEV+IRV deadline, days — (current LEV+IRV)		
	Class A/B	Class C	Class D
N5	2 (4)	1 (2)	0.5 (0.5)
N4	4 (8)	2 (4)	1 (2)
N3	16 (32)	8 (16)	4 (8)
N2	48 (96)	24 (48)	12 (24)

Table 4: Proposed VLEV+IRV remediation floor by Class (current LEV+IRV in parentheses); days, 0.5 = 12 hours. The new LEV band (0.30–0.75 EPSS) inherits today’s LEV deadlines; NLEV (≤ 0.30) is unchanged; the NIRV variants relax by the same factor as today. PAIN-1 still carries no FedRAMP deadline.

Net effect: actively-exploited and > 0.75 -EPSS flaws on high-PAIN, internet-reachable assets compress toward *hours* at the stricter Classes, while the broad 0.30–0.75 middle band — previously treated as “not likely” — moves onto a real clock. Mechanically this is additive: the remediation matrix is already a config-driven [Class] [PAIN] [column] lookup, so VLEV is one more column, not a redesign.

16 Conclusion

FedRAMP Rev5 VDR/VER specifies the vocabulary and the axes of vulnerability prioritization but leaves the classifier unspecified. That classifier need not be invented: the CVSS Environmental metric group already encodes “re-weight impact by what is at stake,” which is precisely the agency-impact question. By assigning each asset its CR/IR/AR — via archetypes or any other systematic scheme the CSP owns — and evaluating the two VER axes, a provider obtains a deterministic, auditable, and standards-grounded PAIN and remediation deadline for every finding. We offer the example archetype catalog not as a standard but as an existence proof; we encourage each CSP to publish its own.

A Example archetype catalog

Illustrative only (see Section 10).

Archetype	Lens	CR	IR	AR	Typical members
cicd-pipeline	control	H	H	H	build/deploy runners, artifact signing, registries
orchestrator	control	H	H	H	control plane, etcd, scheduler, coordination
config-actuation	control	H	H	H	IaC/GitOps, schema registry, admin/migration
identity-secrets	control	H	H	H	IdP/SSO, KMS, secrets managers, session stores
security-tooling	control	H	H	M	scanners, SIEM, EDR, runtime security
change-record	control	M	M	M	ITSM/ticketing (record only)
platform-foundation	control	L	H	H	DNS, NTP, service discovery, L4 internal LBs (metadata only)
data-sensitive	data	H	H	H	PII/CUI datastores
data-backbone	data	H	H	H	queues, brokers, the system-of-record DB
app-tier	data	M	M	H	stateless services, APIs, UIs, caches
batch-analytics	data	M	M	L	ETL, reporting, analytics jobs
public-edge	data	L	L	H	load balancers, public web, ingress
internal-tooling	data	L	L	L	dashboards, metrics/log agents
dev-test	data	L	L	L	non-production
unclassified	—	H	H	H	fail-safe default for untagged assets

B Remediation deadline matrices

Days; 0.5 = 12 hours. Columns in order LEV+IRV / LEV+NIRV / NLEV. PAIN-1 has no FedRAMP deadline (any class). Values illustrative of a representative VDR-TFR-PVR profile.

PAIN	Class A / B			Class C			Class D		
	L+I	L+N	NLEV	L+I	L+N	NLEV	L+I	L+N	NLEV
N5	4	8	32	2	4	16	0.5	1	8
N4	8	32	64	4	8	64	2	8	32
N3	32	64	192	16	32	128	8	16	64
N2	96	160	192	48	128	192	24	96	192

C Explicit design opinions (calibratable choices)

IN PLAIN TERMS

Everything in this section is a *judgment call* we made — not something FedRAMP or CVSS dictates. We collect them in one place so reviewers can challenge each one, and so any adopter knows exactly which dials are theirs to turn. None of these is load-bearing for the *method*; they are the parameters the method runs on.

The method deliberately separates standardized arithmetic (Section 6) from a small set of explicit opinions. Each item below states the value we use, why, and the fact that an implementation MAY change it.

1. **PAIN word thresholds** $(\theta_1, \theta_2, \theta_3) = (0.25, 0.55, 0.80)$. The cut points that turn the impact scalar S into Minimal/Narrow/Disruptive/Debilitating (Eq. 3). This is the single most consequential opinion — it sets where a finding crosses an N-level boundary. The values should reproduce senior-analyst triage on a back-tested sample; an adopter SHOULD re-calibrate against its own rated findings and MUST treat the result as governed configuration.
2. **EPSS Likely-Exploitable threshold** $\theta_{\text{epss}} = 0.70$. The EPSS probability at or above which a finding is treated as LEV (Section 8). Higher means fewer findings on the fast track; FedRAMP leaves the framework to the provider.
3. **LEV is a union, not a blend**. A finding is Likely Exploitable if $\text{EPSS} \geq \theta_{\text{epss}}$ *or* exploitation is observed “active” — whichever fires, with no weighting between them.
4. **Internet-reachability counts edge authentication as a mitigation**. We define IRV as “fronted by a public load balancer/route whose edge does not enforce authentication.” A route gated by a native identity-aware proxy (or equivalent) is treated as *not* internet-reachable, on the view that an unauthenticated attacker cannot reach the vulnerable surface. An adopter wanting a stricter posture MAY count any public route as IRV regardless of edge authentication.
5. **Impact-only severity, normalized by 0.915**. We take CVSS’s *Modified Impact Sub-Score* and rescale it to $[0, 1]$ (Eq. 2) rather than computing the full 0–10 environmental score. PAIN is about *impact on the asset*, so we intentionally drop the exploitability half of the base score (and the 6.42 coefficient) from the severity term; exploitability re-enters separately as the LEV/IRV remediation columns.
6. **Technical impact is a floor, not the primary signal**. “technical impact = total” raises each *in-scope* CVSS dimension to High; it never invents impact on a dimension the CVE marks None, and “partial”/absent leaves the vector unchanged. (An earlier draft made technical impact primary with CVSS as the fallback; we reversed that.)
7. **CISA Vulnrichment as the authoritative enrichment source**. The exploitation state (active / PoC / none) and the technical impact (total / partial) are taken from *CISA Vulnrichment*, CISA’s Authorized Data Publisher (ADP) enrichment of CVE records. Deferring to an ADP’s assessment — rather than a bespoke or purely commercial judgment — keeps these inputs authoritative and governed; an adopter MAY substitute or augment with another source (a vendor SSVC decision, a commercial threat-intelligence feed, or the CISA KEV catalog). EPSS, by contrast, supplies the statistical exploitation *likelihood* (item 2); the two are complementary.
8. **Scope is purely hierarchical**. An asset is multi-agency only if it — or its namespace, or the cluster default — is tagged so (most-specific wins); there is no automatic per-archetype escalation. (An earlier draft auto-escalated “shared-infrastructure” archetypes to multi-agency; we removed that in favor of explicit, operator-controlled tagging.)
9. **“Archetype”, not “type”**. We classify assets by their *role/usage pattern*, not their intrinsic kind, and the label reflects that: the same software lands in different archetypes by how it is used (an in-memory store is **app-tier** as a cache, **identity-secrets** as a session store, **data-backbone** as a broker). “Type” would wrongly suggest an intrinsic property of the software and collides with overloaded terms (e.g. Kubernetes Service **type**); “archetype” names the pattern an asset matches in the estate. An adopter MAY rename the label, but the role-not-kind semantics should be preserved.

10. **Fail-safe defaults to conservative High.** An unclassified asset defaults to CR/IR/AR all High so it scores loudly, rather than to a low default. “Unknown” is treated as serious until classified.
11. **CVSS v3.1, not v4.0.** We build on v3.1’s closed-form environmental formula; v4.0 findings are approximated by mapping VC/VI/VA onto *C/I/A* (Section 13).
12. **The archetype catalog and its CR/IR/AR values (Appendix A).** Illustrative only — each CSP SHOULD derive its own from its system categorization.
13. **The remediation deadline values (Appendix B).** Representative of a plausible VDR-TFR-PVR profile; the authoritative day counts follow FedRAMP and the provider’s Certification-Class commitments.